TITLE:

Student Performance Prediction using Machine Learning

1. Introduction

In today's education system, students often struggle to understand their academic performance. Many students are unaware of whether they are likely to pass or fail. This project aims to predict student performance using machine learning techniques based on simple input features.

2. Problem Statement :

The goal of this project is to predict:

Whether a student will pass or fail

The expected marks of the student

This helps in identifying weak students early and improving their performance.

3. Dataset :

The dataset used in this project is manually created and contains the following features:

Study Hours

Attendance (%)

Previous Marks

Assignment Completion (0 or 1)

Sleep Hours

Target variables:

Final Result (Pass/Fail)

Final Marks

4. Methodology :

Data Preprocessing:

Converted categorical values into numeric

Checked for missing values

Structured dataset properly

Models Used:

Logistic Regression → for classification (Pass/Fail)

Linear Regression → for marks prediction

Tools:

Python

Pandas

Scikit-learn

5. Results :

The classification model was able to predict pass/fail with reasonable accuracy.

The regression model predicted marks based on input features.

Study hours and attendance had strong impact on results.

6. Challenges Faced :

Creating dataset manually

Choosing correct features

Understanding model outputs

7. Conclusion :

This project shows how machine learning can be used in education to predict student performance. Even with simple data, useful predictions can be made.

8. Future Improvements :

Use real student dataset

Improve accuracy with more data

Add user interface